

Sprint 5 – Implementing Core Business Rules in Apex

Placement Management System – Development Report

Key Deliverables

- Fetch Student and Job details through SOQL queries
- Block repeat submissions from the same student for a given job
- Check eligibility against CGPA, Branch, active backlog count, and application closing date
- Save new Application records through DML insert operations
- Update the Application Status field as processing progresses
- Full Apex test class coverage, with all test methods passing at 94% coverage

Screenshots Included

This version of the report includes screenshots of the ApplicationService class, the ApplicationServiceTest class, the Apex test run results, and the code coverage summary. Further screenshots (Student, Job, and Application records, along with Execute Anonymous output) can be added on the closing pages prior to final submission.

```

33 if (student.Branch__c != job.Eligible_Branch__c) {
34     return 'Application Rejected. Branch not eligible.';
35 }
36
37 if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38     return 'Application Rejected. Too many active backlogs.';
39 }
40
41 if (Date.today() > job.Closing_Date__c) {
42     return 'Application Rejected. Application deadline is over.';
43 }
44
45 Application__c application = new Application__c();
46
47 application.Student__c = studentId;
48 application.Job__c = jobId;
49 application.Application_Date__c = Date.today();
50 application.Status__c = 'Applied';
51
52 try {
53     insert application;
54
55     return 'Application Submitted Successfully';
56 } catch (Exception e) {
57     return e.getMessage();
58 }
59
60 }
61
62 }
63
64
65 public static String updateApplicationStatus(Id applicationId, String status) {
66
67     Application__c app = [
68         SELECT W.Status__c
69         FROM Application__c
70         WHERE Id = :applicationId
71         LIMIT 1
72     ];
73
74     app.Status__c = status;
75
76     update app;
77
78     return 'Application Status Updated Successfully';
79 }
80
81 }

```



SETUP

Apex Classes

```

1  @isTest
2  public class ApplicationServiceTest {
3
4      @isTest
5      static void testSubmitApplication() {
6
7          // Create Student
8          Student__c student = new Student__c(
9              Name = 'Rahul',
10             Email__c = 'rahul@gmail.com',
11             CGPA__c = 8.5,
12             Branch__c = 'CSE',
13             Active_Backlogs__c = 0,
14             Graduation_Year__c = 2027
15         );
16         insert student;
17
18         // Create Job
19         Job__c job = new Job__c(
20             Name = 'Software Engineer',
21             Company__c = 'TCS',
22             Minimum_CGPA__c = 7.5,
23             Eligible_Branch__c = 'CSE',
24             Backlog_Limit__c = 0,
25             Closing_Date__c = Date.today().addDays(10)
26         );
27         insert job;
28
29         Test.startTest();
30
31         String result =
32             ApplicationService.submitApplication(student.Id, job.Id);
33
34         Test.stopTest();
35
36         System.assertEquals(
37             'Application Submitted Successfully',
38             result
39         );
40     }
41
42     @isTest
43     static void testDuplicateApplication() {
44
45         Student__c student = new Student__c(
46             Name = 'Student1',
47             Email__c = 'student1@gmail.com',
48             CGPA__c = 8,
49             Branch__c = 'CSE',
50             Active_Backlogs__c = 0,
51             Graduation_Year__c = 2027
52         );
53         insert student;
54
55         Job__c job = new Job__c(
56             Name = 'Developer',
57             Company__c = 'Infosys',
58             Minimum_CGPA__c = 7,
59             Eligible_Branch__c = 'CSE',
60             Backlog_Limit__c = 0,
61             Closing_Date__c = Date.today().addDays(5)
62         );
63         insert job;
64
65         ApplicationService.submitApplication(student.Id, job.Id);
66
67         String result =

```

```

67 string result =
68     ApplicationService.submitApplication(student.Id, job.Id);
69
70     System.assert(
71         result.contains("Duplicate")
72     );
73 }
74 @isTest
75 static void testLowCGPA() {
76
77     Student__c student = new Student__c(
78         Name="Low CGPA",
79         Email__c="low@test.com",
80         CGPA__c=6.0,
81         Branch__c="CSE",
82         Active_Backlogs__c=0,
83         Graduation_Year__c=2027
84     );
85     insert student;
86
87     Job__c job = new Job__c(
88         Name="Developer",
89         Company__c="Infosys",
90         Minimum_CGPA__c=7.5,
91         Eligible_Branch__c="CSE",
92         Backlog_Limit__c=0,
93         Closing_Date__c=Date.today().addDays(5)
94     );
95     insert job;
96
97     String result =
98         ApplicationService.submitApplication(student.Id, job.Id);
99
100     System.assert(result.contains("CGPA"));
101 }
102 @isTest
103 static void testBranchValidation() {
104
105     Student__c student = new Student__c(
106         Name="Branch Test",
107         Email__c="branch@test.com",
108         CGPA__c=9,
109         Branch__c="ECE",
110         Active_Backlogs__c=0,
111         Graduation_Year__c=2027
112     );
113     insert student;
114
115     Job__c job = new Job__c(
116         Name="Developer",
117         Company__c="Infosys",
118         Minimum_CGPA__c=7,
119         Eligible_Branch__c="CSE",
120         Backlog_Limit__c=0,
121         Closing_Date__c=Date.today().addDays(5)
122     );
123     insert job;
124
125     String result =
126         ApplicationService.submitApplication(student.Id, job.Id);
127
128     System.assert(result.contains("Branch"));
129 }
130 @isTest
131 static void testBacklogValidation() {
132
133     Student__c student = new Student__c(
134         Name="Backlog Test",

```

Apex Classes

```

145 // Job__c job = new Job__c(
146 //     Name="Developer",
147 //     Company__c="Infosys",
148 //     Minimum_CGPA__c=7,
149 //     Eligible_Branch__c="CSE",
150 //     Backlog_Limit__c=0,
151 //     Closing_Date__c=Date.today().addDays(5)
152 // );
153 // insert job;
154
155 // String result =
156 //     ApplicationService.submitApplication(student.Id, job.Id);
157 // System.assert(result.contains("Duplicate"));
158
159 @isTest
160 static void testBranchValidation() {
161
162     Student__c student = new Student__c(
163         Name="Branch Test",
164         Email__c="branch@test.com",
165         CGPA__c=9,
166         Branch__c="ECE",
167         Active_Backlogs__c=0,
168         Graduation_Year__c=2027
169     );
170     insert student;
171
172     Job__c job = new Job__c(
173         Name="Developer",
174         Company__c="Infosys",
175         Minimum_CGPA__c=7,
176         Eligible_Branch__c="CSE",
177         Backlog_Limit__c=0,
178         Closing_Date__c=Date.today().addDays(5)
179     );
180     insert job;
181
182     String result =
183         ApplicationService.submitApplication(student.Id, job.Id);
184     System.assert(result.contains("Branch"));
185 }
186
187 @isTest
188 static void testBacklogValidation() {
189
190     Student__c student = new Student__c(
191         Name="Backlog Test",

```

Application Test Execution

To run one or more Apex or flow unit tests, click Select Tests. To view the current Apex code coverage for an individual class or your organization, go to the [Apex Classes](#) page.

 In Winter '26 and later, this page shows test details for multiple categories of tests, including Apex and flow tests. For more info about how to [Test Your Changes](#), go to Salesforce Help.

[Select Tests...](#) [Developer Console](#) [Options...](#) [View Test History](#)

Abort			
☐	Status	Class	Result
Test Run: 2026-08-03 11:59:46, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☐		[View] ApplicationServiceTest	
Test Run: 2026-08-03 11:34:33, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(7/7) Test Methods Passed
Test Run: 2026-08-03 11:34:26, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(7/7) Test Methods Passed
Test Run: 2026-08-03 11:30:10, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(2/2) Test Methods Passed
Test Run: 2026-08-03 11:29:53, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(2/2) Test Methods Passed

Code coverage



Apex Class
ApplicationService

[Back to List](#)

Apex Class Detail

[Edit](#) [Delete](#) [Download](#) [Security](#) [Show Dependencies](#)

Name	ApplicationService	Status	Active
Namespace Prefix		Code Coverage	94% (32/34)
Created By	Java Sri Padilam, 03/08/2026, 10:33 am	Last Modified By	Java Sri Padilam, 03/08/2026, 11:54 am

[Class Body](#) [Class Summary](#) [Version Settings](#) [Trace Flags](#)

Application Record

Student

Swathi

New ContactEditNew Opportunity

Related

Details

Student Name

Swathi

Owner

Jaya Sri Padilam

Email

swathi@gmail.com

CGPA

8.50

Branch

CSE

Active Backlogs

0

Graduation Year

2027

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Job record

Job

Software Engineer

New ContactEditNew Opportunity

Related

Details

Job Title

Software Engineer

Owner

Jaya Sri Padilam

Company

TCS

Minimum CGPA

7.50

Closing Date

31/06/2026

Eligible Branch

CSE

Backlog Limit

0

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Execute Anonymous output

The screenshot shows the Salesforce Developer Console interface. The top bar indicates the browser is Google Chrome and the URL is `resourceful-narwhal-rffxm7-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. The tabs show the current file is `AppointmentFuture.apxc` and the active log is `Log executeAnonymous @8/3/2026, 11:45:05 AM`.

The **Execution Log** section displays a single log entry:

Timestamp	Event	Details
11:45:05:059	USER_DEBUG	[4]DEBUG Duplicate Application! Student already applied.

Below the log, there is a filter bar with options: ☐ This Frame, ☐ Executable, ☒ Debug Only, and ☐ Filter. A text input field says "Click here to filter the log".

The **Logs** section is active, showing a table of log entries:

User	Application	Operation	Time	Status	Read	Size
Jaya Sri Padilam	Unknown	/services/data/v67.0/t...	8/3/2026, 11:45:05 AM	Success		6.87 KB

Github Repository

<https://github.com/7jayasri/Placement-Management-System>